Score Function

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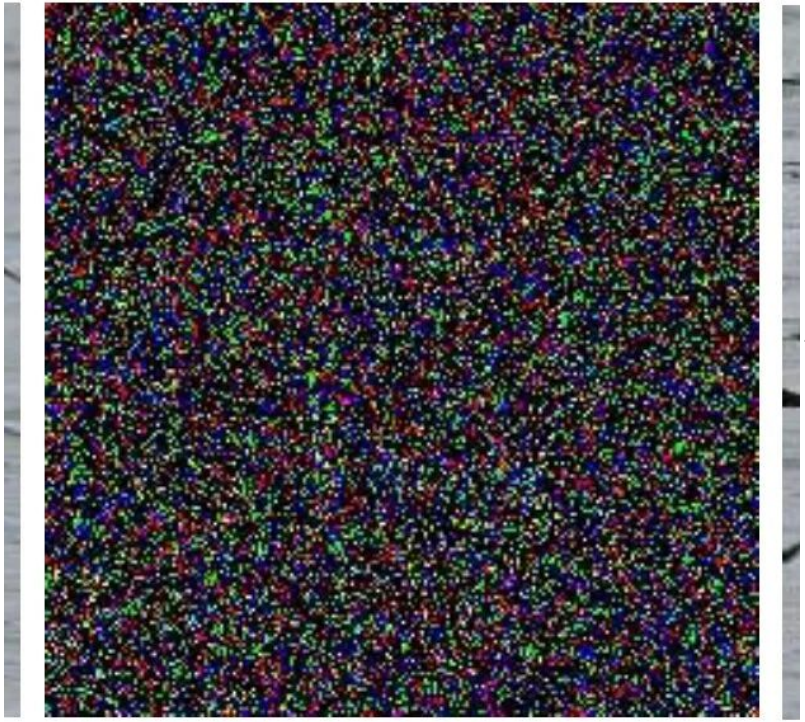


(b) Gaussian noise (GN)

https://www.researchgate.net/profile/Cheng-Lin-Liu-4/publication/347669614/figure/fig4/AS:11431281094995099@1667689576182/Normal-image-a-and-generated-noises-Gaussian-noise-b-and-synthetic-noise-c-on_Q640.jpg

Generated by ChatGPT

Reverse Process

From Last Week: (Anderson's Theorem)

Forward SDE:

$$\frac{dx}{dt} = -\lambda x + \eta(t)$$

Corresponding Backward SDE (Anderson's Theorem):

$$\frac{dx}{d\tau} = [\lambda x + 2D\nabla_x \log p(x,t)] + \eta(\tau)$$

Score Function

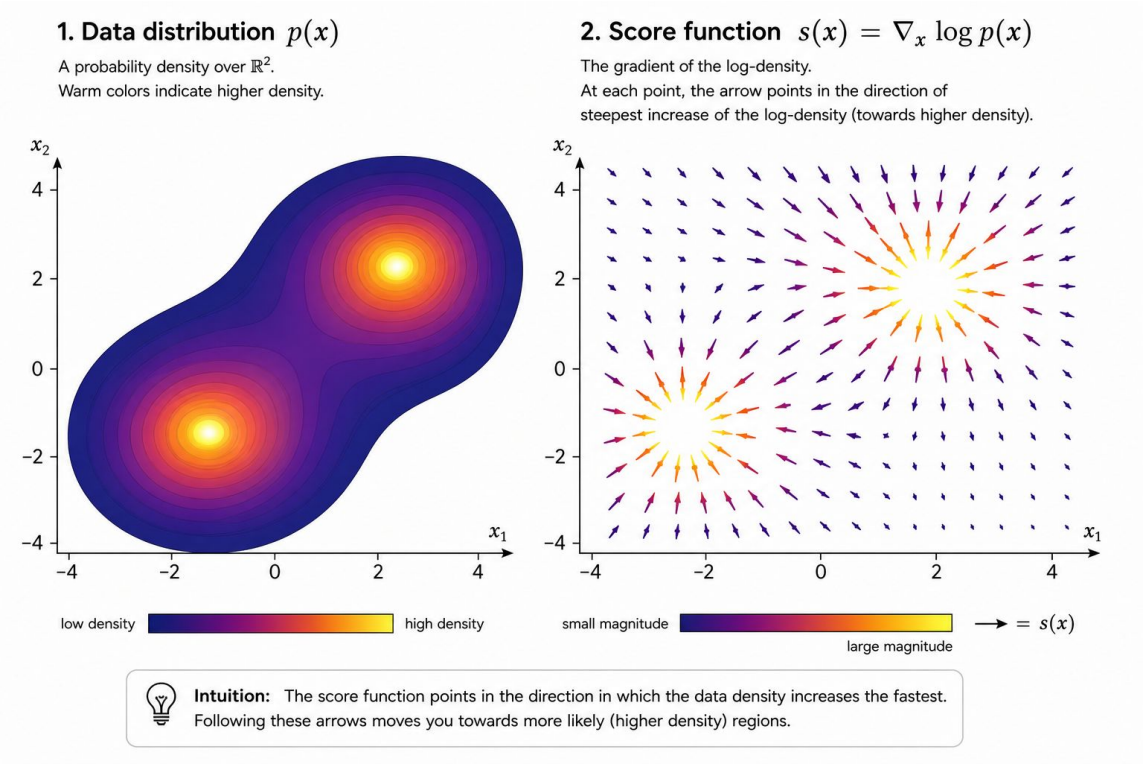
Intuition Score Function

– points in the direction where the data becomes more likely

– follows the direction:

one can recover the data distribution

– extra term is needed !!!



Example:

Initial condition (superposition of delta peaks):

$$p(x,0) = \frac{1}{2}\delta(x+a) + \frac{1}{2}\delta(x-a)$$

Resulting $p(x,t)$ with $m_t = ae^{-\lambda t}$ and $v_t = \frac{D}{\lambda}(1-e^{-2\lambda t})$:

$$p(x,t) = \frac{1}{2\sqrt{2\pi v_t}}\left[\exp\left(-\frac{(x+m_t)^2}{2v_t}\right) + \exp\left(-\frac{(x-m_t)^2}{2v_t}\right)\right]$$

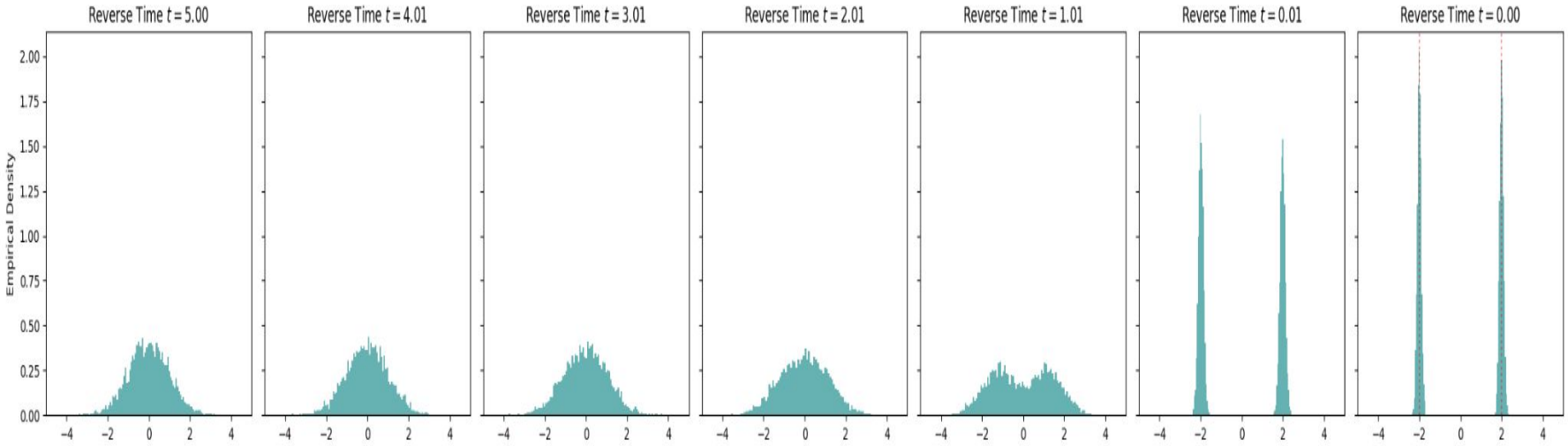
Goal: Calculate the score function $\nabla_x \log p(x, t) = \frac{\partial_x p(x,t)}{p(x,t)}$

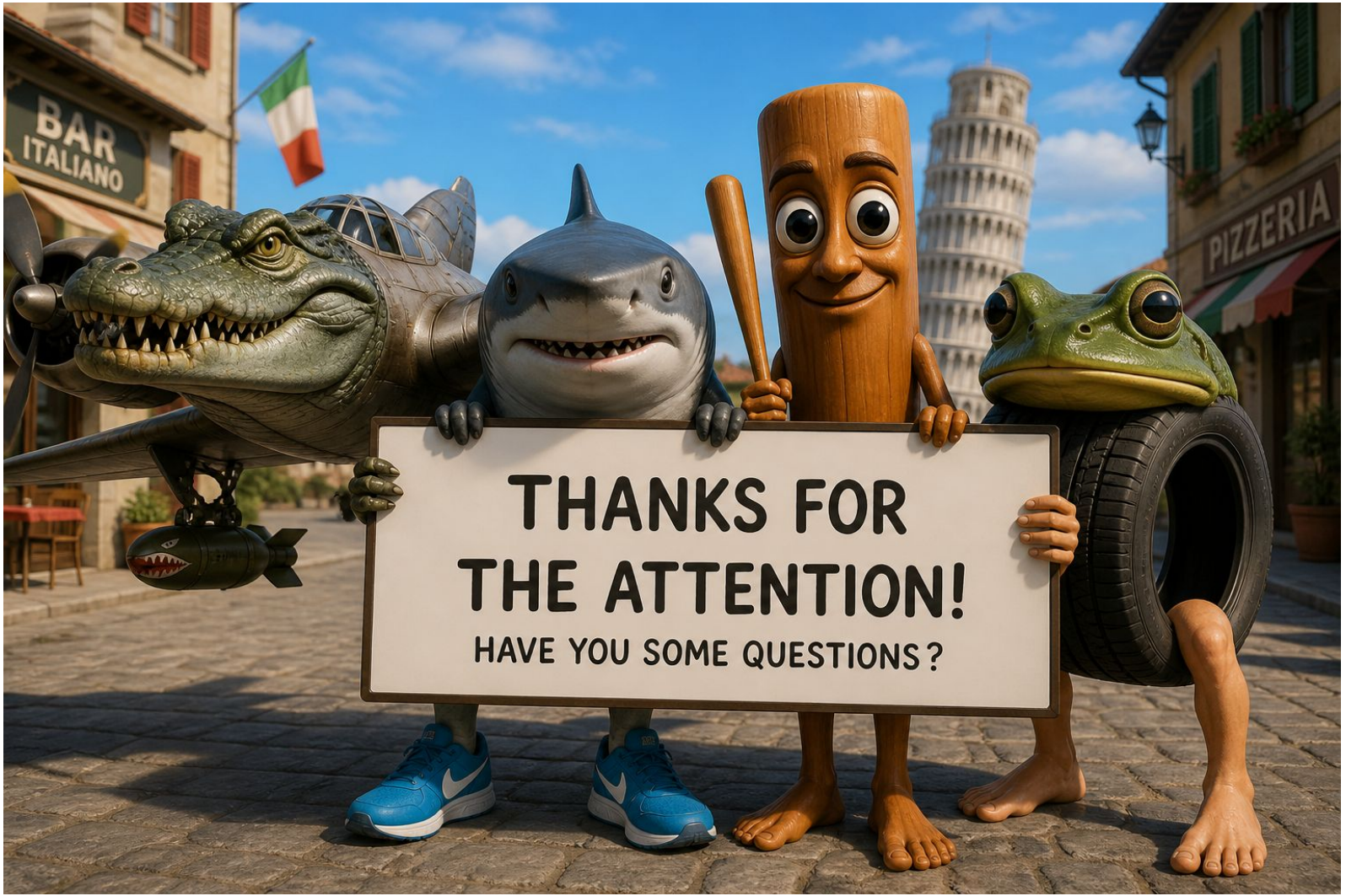
After some Steps ^^

$$\nabla_x \log p(x,t) = -\frac{x}{v_t} + \frac{m_t}{v_t} \tanh\left(\frac{x m_t}{v_t}\right)$$

Fully expanded with original parameters:

$$\nabla_x \log p(x,t) = \frac{-x + a e^{-\lambda t} \tanh\left(\frac{x a e^{-\lambda t}}{\frac{D}{\lambda}(1-e^{-2\lambda t})}\right)}{\frac{D}{\lambda}(1-e^{-2\lambda t})}$$


Reverse Time $t = 5.00$
Reverse Time $t = 4.01$
Reverse Time $t = 3.01$
Reverse Time $t = 2.01$
Reverse Time $t = 1.01$
Reverse Time $t = 0.01$
Reverse Time $t = 0.00$
Empirical Density
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
−4
−2
0
2
4


BAR
ITALIANO
PIZZERIA
THANKS FOR
THE ATTENTION!
HAVE YOU SOME QUESTIONS?